

University of Wah

Department of Computer Science



Submit to:

Ms. Wasifa Kanwal

Submit by:

Muhammad Talha

Course:

Information and Security Lab - CS-251L

Reg No:

UW-24-CS-BS-101

Section:

BS-CS-3rd-B

Lab # 06

```
def manual_zigzag(text, rails):
    grid = [[" " for _ in range(len(text))] for _ in range(rails)]
    row = 0
    direction = 1

    for col, ch in enumerate(text):
        grid[row][col] = ch
        row += direction

        if row == rails:
            row = rails - 2
            direction = -1
        elif row == -1:
            row = 1
            direction = 1

    return grid

def grid_to_cipher(grid):
    result = ""
    for row in grid:
        for ch in row:
            if ch != " ":
                result += ch
    return result
```

```
def manual_decrypt(cipher, rails):
    length = len(cipher)

    (variable) row: Literal[0] length)) for _ in range(rails)]
    row = 0
    direction = 1
    for col in range(length):
        mark[row][col] = "*"
        row += direction

        if row == rails:
            row = rails - 2
            direction = -1
        elif row == -1:
            row = 1
            direction = 1

    print("\n--- Decryption Step A: Marked Pattern ---")
    for r in mark:
        print("".join(r))

    index = 0
```

```

for r in range(rails):
    for c in range(length):
        if mark[r][c] == "*" and index < len(cipher):
            mark[r][c] = cipher[index]
            index += 1

print("\n--- Decryption Step B: Filled Rails ---")
for r in mark:
    print("".join(r))

result = ""
row = 0
direction = 1
for col in range(length):
    result += mark[row][col]
    row += direction
    if row == rails:
        row = rails - 2
        direction = -1
    elif row == -1:
        row = 1
        direction = 1

return result

```

```

def encryptRailFence(text, key):
    rail = [['\n' for _ in range(len(text))] for _ in range(key)]
    dir_down = False
    row, col = 0, 0

    for i in range(len(text)):
        if (row == 0) or (row == key - 1):
            dir_down = not dir_down

        rail[row][col] = text[i]
        col += 1

        if dir_down:
            row += 1
        else:
            row -= 1

    result = []
    for i in range(key):
        for j in range(len(text)):
            if rail[i][j] != '\n':
                result.append(rail[i][j])
    return "".join(result)

```

```

def decryptRailFence(cipher, key):
    rail = [['\n' for _ in range(len(cipher))] for _ in range(key)]
    dir_down = None
    row, col = 0, 0

    # Mark pattern
    for i in range(len(cipher)):
        if row == 0:
            dir_down = True
        if row == key - 1:
            dir_down = False

        rail[row][col] = '*'
        col += 1

        if dir_down:
            row += 1
        else:
            row -= 1

    index = 0
    for i in range(key):
        for j in range(len(cipher)):
            if rail[i][j] == '*' and index < len(cipher):
                rail[i][j] = cipher[index]
                index += 1

```

```

result = []
row, col = 0, 0
for i in range(len(cipher)):
    if row == 0:
        dir_down = True
    if row == key - 1:
        dir_down = False

    if rail[row][col] != '*':
        result.append(rail[row][col])
        col += 1

    if dir_down:
        row += 1
    else:
        row -= 1

    return "".join(result)

plaintext = "Talha Data Structure"
key = 5

```

```
zig = manual_zigzag(plaintext, key)

print("=== MANUAL ZIGZAG PATTERN ===")
for r in zig:
    print("".join(r))

manual_cipher = grid_to_cipher(zig)
print("\nManual Ciphertext:", manual_cipher)

manual_decrypted = manual_decrypt(manual_cipher, key)
print("\nManual Decrypted Text:", manual_decrypted)

code_cipher = encryptRailFence(plaintext, key)
code_decrypted = decryptRailFence(code_cipher, key)

print("\n=== CODE OUTPUTS ===")
print("Code Ciphertext:", code_cipher)
print("Code Decrypted:", code_decrypted)

print("\nMatch Ciphertext? ", manual_cipher == code_cipher)
print("Match Decryption? ", manual_decrypted == plaintext)
```

Output

```
=== MANUAL ZIGZAG PATTERN ===
T      t      t
a      a a     c u
l  D      u  r
h      S r     e
      a      t

Manual Ciphertext: TttaaacuIDurhSreat

--- Decryption Step A: Marked Pattern ---
*      *      *
*      * *     * *
*      *      *
* *      * *
      *      *

--- Decryption Step B: Filled Rails ---
T      t      t
a      a a     c u
l  D      u  r
h S      r e
      a      t

Manual Decrypted Text: TalhaSDataurterctu

=== CODE OUTPUTS ===
Code Ciphertext: TttaaacuID urh Sreat
Code Decrypted: Talha Data Structure

Match Ciphertext? False
Match Decryption? False
```